Some Results of Antibandwidth on Generalized Prism Graphs*

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Abstract

The antibandwidth problem consists of placing the vertices of a graph on a line in consecutive integer points in such a way that the minimum difference of adjacent vertices is maximized. The generalized prism graph is the graph defined by the graph Cartesian product $P_m \times C_n$, where P_m denotes the path on m vertices and C_n denotes the cycle on n vertices. In this paper, we investigate the antibandwidth problem on generalized prism graphs and obtain the following results:

(1) *For even n, if $m \geqslant n$, then* $\mathrm{ab}(P_m \times C_n) = \frac{n(m-1)}{2}$*; otherwise,* $\mathrm{ab}(P_m \times C_n) \geqslant \frac{n(m-1)}{2}$.

(2) *For odd n, if $m \leqslant n$, then* $\mathrm{ab}(P_m \times C_n) = \frac{m(n-1)}{2}$*; otherwise,* $\mathrm{ab}(P_m \times C_n) \geqslant \frac{m(n-1)}{2}$.

(3) *If n is even,* $\mathrm{ab}(P_2 \times C_n) = n-2$*; otherwise,* $\mathrm{ab}(P_2 \times C_n) = n-1$.

Keyword: antibandwidth, Cartesian product, generalized prism graphs

1. Introduction

Let $[a, b]$ denote the set of integers x with $a \leqslant x \leqslant b$. By a *labeling* for a graph $G = (V, E)$ on n vertices we mean a bijection $f : V \to [1, n]$. The *antibandwidth of G with respect to f* is defined as

$$\mathrm{ab}(G, f) = \min_{uv \in E} |f(u) - f(v)|.$$

The *antibandwidth* of a graph G is the maximum antibandwidth of G over all labelings f, i.e.,

$$\mathrm{ab}(G) = \max_f \mathrm{ab}(G, f).$$

The problem of finding antibandwidth in a graph was originally studied in [8] under the term *separation number* in a connection with multiprocessor scheduling problems. In addition, antibandwidth is sometimes called the *dual bandwidth* [9] since it can be viewed as dual to the well-known *bandwidth problem* [3] in which the maximum difference between labels of adjacent vertices in the linear layout is minimized. Finding antibandwidth is naturally motivated by some real-life applications such as obnoxious facility location problems [2], radio-colouring [5], and several task or game scheduling problems [8].

Leung *et al.* [8] showed that finding antibandwidth in a graph is NP-hard. Some basic relations between $\mathrm{ab}(G)$ and other graph parameters have been derived in [8–10]. So far the only known graph classes possessed polynomial time algorithms for finding antibandwidth are the complements of interval graphs [6], arborescent comparability graphs and threshold graphs [4]. Also, exact results and tight bounds are known for paths, cycles, special trees [1, 7, 14], meshes [11, 12], and hypercubes [11, 13].

In this paper, we investigate the antibandwidth problem on a family of graphs called generalized prism graphs, denoted by $P_m \times C_n$. Note that a generalized prism graph is a supergraph of a mesh $P_m \times P_n$ and a subgraph of a torus $C_m \times C_n$. Since the exact results of antibandwidth are known for meshes and unknown for tori, it is of significance to study the antibandwidth problem on generalized prism graphs. According to the parity of n, we provide different labeling schemes to obtain lower bounds of $\mathrm{ab}(P_m \times C_n)$. In particular, if n is even and $m \geqslant n$, we acquire the exact value $\mathrm{ab}(P_m \times C_n) = \frac{n(m-1)}{2}$. By contrast, if n is odd and $m \leqslant n$, we acquire the exact value $\mathrm{ab}(P_m \times C_n) = \frac{m(n-1)}{2}$. Moreover, we prove that $\mathrm{ab}(P_2 \times C_n) = n - 2$ if n is even, and $\mathrm{ab}(P_2 \times C_n) = n - 1$ if n is odd.

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2. Preliminaries

Let P_n denotes the n-vertex path where $V(P_n) = \{v_1, v_2, \ldots, v_n\}$ and $E(P_n) = \{v_i v_{i+1} : i = 1, 2, \ldots n-1\}$. Similarly, let C_n denotes the n-vertex cycle where $V(C_n) = \{v_1, v_2, \ldots, v_n\}$ and $E(C_n) = \{v_i v_{i+1} : i = 1, 2, \ldots n-1\} \cup \{v_n v_1\}$.

The *Cartesian product* of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, denoted by $G_1 \times G_2$, is the graph with $V_1 \times V_2$ as its vertex set and an edge $(u_1, u_2)(v_1, v_2) \in E(G_1 \times G_2)$ if and only if either $u_1 v_1 \in E_1$ and $u_2 = v_2$ or $u_2 v_2 \in E_2$ and $u_1 = v_1$. For notational convenience, if $V_1 = \{u_1, u_2, \ldots, u_m\}$ and $V_2 = \{w_1, w_2, \ldots, w_n\}$, we simply write a vertex $v_{i,j}$ instead of (u_i, w_j) in $P_n \times C_n$ for $1 \leqslant i \leqslant m$ and $1 \leqslant j \leqslant n$. Many well-known graph classes are defined by Cartesian products, e.g., $P_m \times P_n$ is called a *mesh*, $C_m \times C_n$ is called a *torus* (or *toroidal mesh*), $P_2 \times C_n$ is called a *prism graph* (or a *circular ladder graph*), and $P_m \times C_n$ is called a *generalized prism graph.*

Miller and Pritikin [10] proved that $\lceil \frac{n(m-1)}{2} \rceil \leqslant \mathrm{ab}(P_m \times P_n) \leqslant \lfloor \frac{mn}{2} \rfloor$ for $m \times n$ meshes with $m \geqslant n$. The lower bound is obtained by their general labeling scheme applicable to bipartite graphs and the upper bound comes from a simple observation that $\mathrm{ab}(G) \leqslant \lfloor \frac{|V(G)|}{2} \rfloor$ for graphs G without isolated vertices. Raspaud *et al.* [11] recently developed an upper bound method suitable for bipartite graphs and obtained the following result.

Theorem 1. [11] *If $m \geqslant n$, then* $\mathrm{ab}(P_m \times P_n) = \left\lceil \frac{n(m-1)}{2} \right\rceil$.

Raspaud *et al.* [11] also acquired the exact value of antibandwidth for square tori $C_n \times C_n$.

Theorem 2. [11]

$$\mathrm{ab}(C_n \times C_n) = \begin{cases} \frac{(n-2)n}{2}, & \text{if } n \text{ is even} \\ \frac{(n-2)(n+1)}{2}, & \text{if } n \text{ is odd.} \end{cases}$$

For a graph G, a subgraph H of G is called a *spanning subgraph* if $V(H) = V(G)$ and $E(H) \subseteq E(G)$. A simple observation shows that if H is a spanning subgraph of a graph G, then $\mathrm{ab}(G) \leqslant \mathrm{ab}(H)$. The following proposition are useful when we study upper bounds of antibandwidth for tori and generalized prism graphs.

Proposition 1. $\mathrm{ab}(C_m \times C_n) \leqslant \mathrm{ab}(P_m \times C_n) \leqslant \mathrm{ab}(P_m \times P_n)$.

3. General lower bounds of $\mathrm{ab}(P_m \times C_n)$

We start this section by developing general lower bounds for $\mathrm{ab}(P_m \times C_n)$. Note that $P_m \times C_n$ is a bipartite graph provided n is even. In this case, a lower bound of $\mathrm{ab}(P_m \times C_n)$ can be obtained by a labeling scheme used for achieving a lower bound of $\mathrm{cab}(P_m \times P_n)$ in [11], where $\mathrm{cab}(G)$ is the *cyclic antibandwidth* of a graph G (i.e., the vertices of G are mapped bijectively into a cycle of length $|V(G)|$ such that the minimal distance, measured in the cycle, of adjacent vertices is maximized). On the other hand, for odd n, we provide another labeling scheme to obtain a lower bound of $\mathrm{ab}(P_m \times C_n)$.

Lemma 1. *For even n,* $\mathrm{ab}(P_m \times C_n) \geqslant \frac{n(m-1)}{2}$.

Proof. For each vertex $v_{i,j}$, the label $f(v_{i,j})$ is defined as

$$f(v_{i,j}) = \begin{cases} (i-1) \cdot \frac{n}{2} + \lceil \frac{j}{2} \rceil, & \text{if } i+j \text{ is even} \\ (m+i-1) \cdot \frac{n}{2} + \lceil \frac{j}{2} \rceil, & \text{otherwise.} \end{cases}$$

The labeling f was defined equivalently in [11, Theorem 4.2]. For example, see Table 1 for the labeling of $m = 5$ and $n = 6$. Let $\alpha = n(m-1)/2$. In [11], Raspaud *et al.* showed the following results.

- For $1 \leqslant i < m$ and $1 \leqslant j \leqslant n$, $|f(v_{i,j}) - f(v_{i+1,j})| \geqslant \alpha$;
- For $1 \leqslant i \leqslant m$ and $1 \leqslant j < n$, $|f(v_{i,j}) - f(v_{i,j+1})| \geqslant \alpha$.

To prove the lemma, it remains to show that $|f(v_{i,n}) - f(v_{i,1})| \geqslant \alpha$ for all $i \in [1, m]$. The inequality can be easily verified as follows. If i is even, then

$$\begin{aligned} & |f(v_{i,n}) - f(v_{i,1})| \\ = \; & f(v_{i,1}) - f(v_{i,n}) \\ = \; & (m+i-1) \cdot \frac{n}{2} + \left\lceil \frac{1}{2} \right\rceil - (i-1) \cdot \frac{n}{2} - \left\lceil \frac{n}{2} \right\rceil \\ = \; & \frac{mn}{2} + 1 - \frac{n}{2} \\ > \; & \frac{n(m-1)}{2}. \end{aligned} \tag{1}$$

Otherwise,

$$\begin{aligned} & |f(v_{i,n}) - f(v_{i,1})| \\ = \; & f(v_{i,n}) - f(v_{i,1}) \\ = \; & (m+i-1) \cdot \frac{n}{2} + \left\lceil \frac{n}{2} \right\rceil - (i-1) \cdot \frac{n}{2} - \left\lceil \frac{1}{2} \right\rceil \\ = \; & \frac{mn}{2} + \frac{n}{2} - 1 \\ > \; & \frac{n(m-1)}{2}. \end{aligned} \tag{2}$$

By (1) and (2), the lemma follows. □

Table 1: The labeling of the vertices of $P_5 \times C_6$ applied in Lemma 1

1	16	2	17	3	18
19	4	20	5	21	6
7	22	8	23	9	24
25	10	26	11	27	12
13	28	14	29	15	30

Lemma 2. *For odd n,* $\mathrm{ab}(P_m \times C_n) \geqslant \frac{m(n-1)}{2}$.

Proof. For each vertex $v_{i,j}$, the label $f(v_{i,j})$ is defined as

$$f(v_{i,j}) = \begin{cases} i+m\cdot\left(\frac{i(n-1)-j+1}{2} \mod n\right), & \text{if } j \text{ is odd} \\ i+m\cdot\left(\frac{n+i(n-1)-j+1}{2} \mod n\right), & \text{if } j \text{ is even.}\end{cases}$$

For example, see Table 2 for the labeling of $m = 4$ and $n = 7$. Let $\beta = m(n-1)/2$. We first show that $|f(v_{i,j}) - f(v_{i,j+1})| \geqslant \beta$ for all $i \in [1, m]$ and $j \in [1, n]$, where the index $j+1$ is taken modulo n:

Case: j is odd. Clearly, $i(n-1) - j + 1$ is even. Let $\Delta = (i(n-1) - j + 1)/2$. Then,

$$\begin{aligned}&|f(v_{i,j}) - f(v_{i,j+1})| \\ =\ & \left|i + m\cdot\left(\frac{i(n-1)-j+1}{2} \mod n\right) - \right. \\ & \left. i - m\cdot\left(\frac{n+i(n-1)-(j+1)+1}{2} \mod n\right)\right| \\ =\ & m\cdot\left|(\Delta \mod n) - \left((\Delta + \frac{n-1}{2}) \mod n\right)\right|.\end{aligned}$$

Thus, if $((\Delta + \frac{n-1}{2}) \bmod n) \geqslant (\Delta \bmod n)$, then $|f(v_{i,j}) - f(v_{i,j+1})| = \frac{m(n-1)}{2}$. Otherwise, $|f(v_{i,j}) - f(v_{i,j+1})| = m\cdot(n - \frac{n-1}{2}) = \frac{m(n+1)}{2} > \frac{m(n-1)}{2}$.

Case: j is even. Clearly, $n + i(n-1) - j + 1$ is even. Let $\Delta = (n + i(n-1) - j + 1)/2$. Then,

$$\begin{aligned}&|f(v_{i,j}) - f(v_{i,j+1})| \\ =\ & \left|i + m\cdot\left(\frac{n+i(n-1)-j+1}{2} \mod n\right) - \right. \\ & \left. i - m\cdot\left(\frac{i(n-1)-(j+1)+1}{2} \mod n\right)\right| \\ =\ & m\cdot\left|(\Delta \mod n) - \left((\Delta - \frac{n+1}{2}) \mod n\right)\right|.\end{aligned}$$

Thus, if $((\Delta + \frac{n+1}{2}) \bmod n) \geqslant (\Delta \bmod n)$, then $|f(v_{i,j}) - f(v_{i,j+1})| = m\cdot(n - \frac{n+1}{2}) = \frac{m(n-1)}{2}$. Otherwise, $|f(v_{i,j}) - f(v_{i,j+1})| = \frac{m(n+1)}{2} > \frac{m(n-1)}{2}$.

Next, we show that $|f(v_{i,j}) - f(v_{i+1,j})| \geqslant \beta$ for all $i \in [1, m-1]$ and $j \in [1, n]$ as follows:

Case: j is odd. Let $\Delta = (i(n-1) - j + 1)/2$. Then,

$$\begin{aligned}&|f(v_{i,j}) - f(v_{i+1,j})| \\ =\ & \left|i + m\cdot\left(\frac{i(n-1)-j+1}{2} \mod n\right) - \right. \\ & \left. (i+1) - m\cdot\left(\frac{(i+1)(n-1)-j+1}{2} \mod n\right)\right| \\ =\ & \left|m\cdot(\Delta \mod n) - m\cdot\left((\Delta + \frac{n-1}{2}) \mod n\right) - 1\right|.\end{aligned}$$

Thus, if $((\Delta + \frac{n-1}{2}) \bmod n) \geqslant (\Delta \bmod n)$, then $|f(v_{i,j}) - f(v_{i+1,j})| = |m\cdot\frac{n-1}{2} + 1| = \frac{m(n-1)}{2} + 1 > \frac{m(n-1)}{2}$. Otherwise, $|f(v_{i,j}) - f(v_{i+1,j})| = |m\cdot(n - \frac{n-1}{2}) - 1| = \frac{m(n+1)}{2} - 1 > \frac{m(n-1)}{2}$.

Case: j is even. Let $\Delta = (n + i(n-1) - j + 1)/2$. Then,

$$\begin{aligned}&|f(v_{i,j}) - f(v_{i+1,j})| \\ =\ & \left|i + m\cdot\left(\frac{n+i(n-1)-j+1}{2} \mod n\right) - \right. \\ & \left. (i+1) - m\cdot\left(\frac{n+(i+1)(n-1)-j+1}{2} \mod n\right)\right| \\ =\ & \left|m\cdot(\Delta \mod n) - m\cdot\left((\Delta + \frac{n-1}{2}) \mod n\right) - 1\right|.\end{aligned}$$

Thus, if $((\Delta + \frac{n-1}{2}) \bmod n) \geqslant (\Delta \bmod n)$, then $|f(v_{i,j}) - f(v_{i+1,j})| = |m\cdot\frac{n-1}{2} + 1| = \frac{m(n-1)}{2} + 1 > \frac{m(n-1)}{2}$. Otherwise, $|f(v_{i,j}) - f(v_{i+1,j})| = |m\cdot(n - \frac{n-1}{2}) - 1| = \frac{m(n+1)}{2} - 1 > \frac{m(n-1)}{2}$.

Consequently, the difference between labels of any two adjacent vertices is at least $\frac{m(n-1)}{2}$. □

Table 2: The labeling of the vertices of $P_4 \times C_7$ applied in Lemma 2

13	25	9	21	5	17	1
26	10	22	6	18	2	14
11	23	7	19	3	15	27
24	8	20	4	16	28	12

According to Theorem 1 and Proposition 1, if $m \geqslant n$, $\mathrm{ab}(P_m \times C_n) \leqslant \lceil\frac{n(m-1)}{2}\rceil$. Otherwise, $\mathrm{ab}(P_m \times C_n) \leqslant \lceil\frac{m(n-1)}{2}\rceil$. Therefore, by Lemma 1 and Lemma 2 we have the following result.

Theorem 3. *The following statements are true:*

(1) If n is even, then $\mathrm{ab}(P_m \times C_n) \geqslant \frac{n(m-1)}{2}$. In particular, the equality holds for $m \geqslant n$.

(2) If n is odd, then $\mathrm{ab}(P_m \times C_n) \geqslant \frac{m(n-1)}{2}$. In particular, the equality holds for $m \leqslant n$.

4. Antibandwidth of $P_2 \times C_n$

In this section, we will compute the exact value of $\mathrm{ab}(P_2 \times C_n)$. Fortunately, by Theorem 3, we know that if n is odd, then $\mathrm{ab}(P_2 \times C_n) = n-1$. However, if n is even, we merely know the lower bound $\mathrm{ab}(P_2 \times C_n) \geqslant \frac{n}{2}$. The following lemma provides an upper bound of $\mathrm{ab}(P_2 \times C_n)$ and shows that the lower bound can be improved to achieve its upper bound.

Lemma 3. *For even n, $\mathrm{ab}(P_2 \times C_n) = n-2$.*

Proof. We prove the lower bound by providing a labeling on the vertices of $P_2 \times C_n$. For each vertex $v_{i,j}$, the label $f(v_{i,j})$ is given by

$$f(v_{i,j}) = \begin{cases} n+j, & \text{if } i+j \text{ is even} \\ (j \bmod n)+1, & \text{otherwise.} \end{cases}$$

For example, if we apply the labeling to $P_2 \times C_8$, then $f(v_{1,j})_{j=1,2,\dots,8} = 9,3,11,5,13,7,15,1$ and $f(v_{2,j})_{j=1,2,\dots,8} = 2,10,4,12,6,14,8,16$.

We now check the difference between labels of adjacent vertices. We first consider two adjacent vertices in the first row. Without loss of generality, we may check $|f(v_{1,j}) - f(v_{1,j-1})|$ and $|f(v_{1,j}) - f(v_{1,j+1})|$ for all even j. If $j \neq n$, then $|f(v_{1,j}) - f(v_{1,j-1})| = |(j+1)-(n+j-1)| = n-2$ and $|f(v_{1,j}) - f(v_{1,j+1})| = |(j+1)-(n+j+1)| = n$. Otherwise, we have $|f(v_{1,n}) - f(v_{1,n-1})| = |1-(2n-1)| = 2n-2$ and $|f(v_{1,n}) - f(v_{1,1})| = |1-(n+1)| = n$. Next, we consider two adjacent vertices in the second row. By symmetry, if j is even, we have $|f(v_{2,j}) - f(v_{2,j-1})| = n$, $|f(v_{2,j}) - f(v_{2,j+1})| = n-2$ if $j \neq n$, and $|f(v_{2,n}) - f(v_{2,1})| = 2n-2$. Finally, we consider two adjacent vertices in the same column. If $j \neq n$, then $|f(v_{1,j}) - f(v_{2,j})| = |(n+j)-(j+1)| = n-1$. Otherwise, $|f(v_{1,n}) - f(v_{2,n})| = |1-2n| = 2n-1$. As a result, the difference between labels of any two adjacent vertices is at least $n-2$.

We proceed the upper bound by contradiction, and so assume that $\mathrm{ab}(P_2 \times C_n) \geqslant n-1$. Let $f : V(P_2 \times C_n) \to [1, 2n]$ be a bijective labeling of the vertices of $P_2 \times C_n$. Without loss of generality, we let $f(v_{1,n}) = n$. By assumption, the three neighbors of $v_{1,n}$ (i.e., $v_{1,1}$, $v_{1,n-1}$ and $v_{2,n}$) must be labeled by 1, $2n-1$ and $2n$. Since $f^{-1}(n) = v_{1,n}$ is the only common neighbor of vertices labeled by 1 and $2n-1$, by symmetry we may assume $f(v_{1,1}) = 1$, $f(v_{1,n-1}) = 2n-1$ and $f(v_{2,n}) = 2n$. Since the three neighbors of $f^{-1}(n+1)$ (respectively, $f^{-1}(n-1)$) are label by 1, 2 and $2n$ (respectively, $2n-2$, $2n-1$ and $2n$), this implies $f(v_{2,1}) = n+1$ and $f(v_{2,2}) = 2$ (respectively, $f(v_{2,n-1}) = n-1$ and $f(v_{2,n-2}) = 2n-2$). In general, we have

$$f(v_{i,j}) = \begin{cases} j, & \text{if } i+j \text{ is even for } j \in [1, \frac{n}{2}-1] \text{ or} \\ & i+j \text{ is odd for } i \in [\frac{n}{2}+1, n-1]; \\ n+j, & \text{if } i+j \text{ is odd for } j \in [1, \frac{n}{2}-1] \\ & \text{or } i+j \text{ is even for } i \in [\frac{n}{2}+1, n-1]. \end{cases} \quad (3)$$

Case: $\frac{n}{2}$ is even. By (3), we have $f(v_{1,\frac{n}{2}-1}) = \frac{n}{2}-1$ and $f(v_{1,\frac{n}{2}+1}) = \frac{3n}{2}+1$. Clearly, the three neighbors of $v_{1,\frac{n}{2}-1}$ are $v_{1,\frac{n}{2}-2}$, $v_{2,\frac{n}{2}-1}$ and $v_{1,\frac{n}{2}}$, and by assumption they must be labeled by $\frac{3n}{2}-2$, $\frac{3n}{2}-1$ and $\frac{3n}{2}$. Again, by (3), we have $f(v_{1,\frac{n}{2}-2}) = \frac{3n}{2}-2$ and $f(v_{2,\frac{n}{2}-1}) = \frac{3n}{2}-1$. This implies that $f(v_{1,\frac{n}{2}}) = \frac{3n}{2}$. Thus, $|f(v_{1,\frac{n}{2}+1}) - f(v_{1,\frac{n}{2}})| = 1 < n-1$, a contradiction.

Case: $\frac{n}{2}$ is odd. By (3), we have $f(v_{1,\frac{n}{2}-1}) = \frac{3n}{2}-1$ and $f(v_{1,\frac{n}{2}+1}) = \frac{n}{2}+1$. Clearly, the three neighbors of $v_{1,\frac{n}{2}-1}$ are $v_{1,\frac{n}{2}-2}$, $v_{2,\frac{n}{2}-1}$ and $v_{1,\frac{n}{2}}$, and by assumption they must be labeled by $\frac{n}{2}-2$, $\frac{n}{2}-1$ and $\frac{n}{2}$. Again, by (3), we have $f(v_{1,\frac{n}{2}-2}) = \frac{n}{2}-2$ and $f(v_{2,\frac{n}{2}-1}) = \frac{n}{2}-1$. This implies that $f(v_{1,\frac{n}{2}}) = \frac{n}{2}$. Thus, $|f(v_{1,\frac{n}{2}+1}) - f(v_{1,\frac{n}{2}})| = 1 < n-1$, a contradiction. □

By Theorem 3 and Lemma 3, we obtain the following result.

Theorem 4. $\mathrm{ab}(P_2 \times C_n) = \begin{cases} n-2, & \textit{if } n \textit{ is even} \\ n-1, & \textit{if } n \textit{ is odd.} \end{cases}$

5. Conclusions

In this paper, we study the antibandwidth problem on generalized prism graphs. Although we have proved exact results under some restricted cases, the problem on generalized prism graphs is still unsolved. Several questions remain open:

- Find exact results of $\mathrm{ab}(P_m \times C_n)$ for even n and $m < n$.
- Find exact results of $\mathrm{ab}(P_m \times C_n)$ for odd n and $m > n$.
- Find exact results of $\mathrm{ab}(C_m \times C_n)$ for $m \neq n$ (i.e., non-square tori).

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